

Question

Given that $K_{a1} = 1.1 \times 10^{-7}$, $K_{a2} = 1.3 \times 10^{-13}$ for H_2S ; $K_{sp}(\text{ZnS}) = 2.5 \times 10^{-22}$. Dissolve 0.01 mol ZnS in 1.0 L hydrochloric acid. The closest value for the minimum concentration of hydrochloric acid required is

A. 0.06 mol/L

B. 0.07 mol/L

C. 0.08 mol/L

D. 0.09 mol/L

E. 0.10 mol/L

F. 0.11 mol/L

Answer

Correct Answer: **E**

Detailed Explanation

This is a very classic physical chemistry problem. The equation for the reaction is $\text{ZnS} + 2\text{H}^+ = \text{Zn}^{2+} + \text{H}_2\text{S}$. It can be calculated that the equilibrium constant for this reaction is:

$$K = \frac{[\text{Zn}^{2+}][\text{H}_2\text{S}]}{[\text{H}^+]^2} = \frac{[\text{Zn}^{2+}][\text{H}_2\text{S}][\text{S}^{2-}]}{[\text{H}^+]^2[\text{S}^{2-}]} = K_{sp}(\text{ZnS}) \frac{[\text{H}_2\text{S}]}{[\text{H}^+]^2[\text{S}^{2-}]} = \frac{K_{sp}(\text{ZnS})}{K_{a1}K_{a2}} = 1.7 \times 10^{-2}$$

CHECKPOINT

1 PTS

$$K = 1.7 \times 10^{-2}$$

When ZnS is completely dissolved, $0.01 \text{ mol} \cdot \text{L} \text{Zn}^{2+}$ and $0.01 \text{ mol} \cdot \text{L} \text{H}_2\text{S}$ are generated. The equilibrium $[\text{H}^+]$ can be calculated:

$$[\text{H}^+]_{eq} = \sqrt{\frac{0.01 \times 0.01}{K}} = 0.077 \text{ mol/L}$$

CHECKPOINT

1 PTS

$$[\text{H}^+]_{eq} = 0.077 \text{ mol/L}$$

Since dissolution occurs, $[\text{H}^+]_{total} = 0.02 + [\text{H}^+]_{eq} = 0.097 \text{ mol/L}$

CHECKPOINT

1 PTS

$$[H^+]_{total} = 0.097 \text{ mol/L}$$